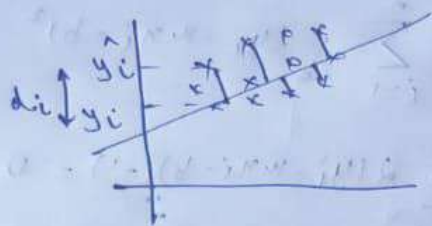


Regression metrics:

MAE → Mean absolute error.



$$AE = |y_1 - \hat{y}_1| + |y_2 - \hat{y}_2| + \dots + |y_n - \hat{y}_n|$$

$$E = \sum_{i=1}^n |y_i - \hat{y}_i|$$

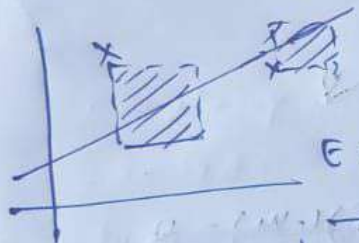
$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Adv → Same unit, robust to outliers.

Disadv → $|x|$ is not diff at $x=0$ (optimization algs cannot be applied)

MSE → Mean squared error
(x^2 is diff at $x=0$) (can be used for optimization)

$$E = (y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + (y_3 - \hat{y}_3)^2 + \dots + (y_n - \hat{y}_n)^2$$



$$E = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Adv → can be used as loss function

Disadv → Diff units (m^2), not robust to outliers

RMSE:

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Adv → Same unit

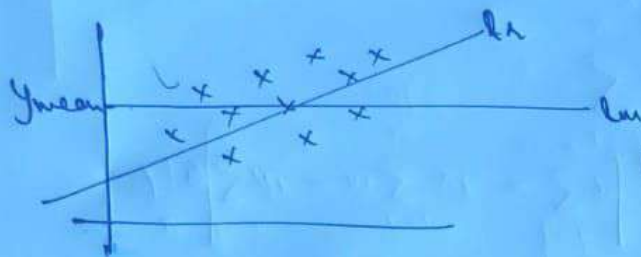
Dis → Not R to outliers.

R^2 score: coefficient of determination, goodness of fit.
(How well data fit in reg model wrt to mean)

$$R^2 = 1 - \frac{SSR}{SSM}$$

Sum of sq error in the regression line

" " " " the mean line,



$$R^2 = 1 - \frac{\left[\sum_{i=1}^n (y_i - \hat{y}_i)^2 \right]_R}{\left[\sum_{i=1}^n (y_i - \bar{y})^2 \right]_M}$$

$$R^2 = 0 \quad (SSR = SSM)$$

(Regression line overlaps on mean line)

(No effect of x axis on data) (useless model)

$$R^2 = 1 \quad (SSR = 0)$$

(No error of regression line) (it passes through all the points)

(Perfect model)

Adjusted R^2 score: Addⁿ of more i/p features helps in the improvement of R^2 score.

$n \rightarrow$ no of rows

$k \rightarrow$ independent vars.

$$R^2_{adj} = 1 - \left[\frac{(1 - R^2)(n - 1)}{(n - 1 - k)} \right]$$